Pilwon Hur

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Appointments

Gwangju Institute of Science and Technology *Gwangju, Korea*
ASSOCIATE PROFESSOR - MECHANICAL ENGINEERING *Dec 2020 - PRESENT*
DIRECTOR - GIST-SAMSUNG INTELLIGENT MOTOR TALENT TRAINING CENTER *Feb 2023 - PRESENT*

- Research Interests: Estimation and Control for Human-Robot Interaction, Free Energy Principle in Human Neuroscience, Rehabilitation Robotics/Engineering for Neurologically-Impaired Patients (e.g., Elderly, Stroke, SCI, MS) or Physically-Disabled Patients (e.g., Amputee), Dynamic Bipedal Walking, Unification of Bipedal Walking, Trajectory Optimization, Prosthetics, Exoskeleton, Sensory Augmentation, Slip/Push Recovery, Virtual/Tele Rehabilitation, Neuromechanics, Biomechanics, Motor Control, Gait, Balance, Equine-Assisted Therapy, Motion Analysis
- Teaching Interests: Robotics, Underactuated Robotics, Nonlinear Control, Geometric Control, Multivariable Control, Optimization, Dynamics, Biomechanics, Sensorimotor System for Human Movement, Estimation Theory, Functional Analysis for Engineering Optimization

Texas A&M University *College Station, TX*
ASSISTANT PROFESSOR - MECHANICAL ENGINEERING *Aug 2014 - Dec 2020*

Texas A&M Engineering Experiment Station *College Station, TX*
CENTER FOR REMOTE HEALTH TECHNOLOGIES AND SYSTEMS *Jun 2015 - Dec 2020*

University of Wisconsin-Milwaukee *Milwaukee, WI*
POSTDOCTORAL RESEARCH FELLOW - CENTER FOR ERGONOMICS *Sep 2010 - Jun 2014*

- Research Interests: Rehabilitation for Stroke, Neuromechanics, Biomechanics, Hand Exoskeleton, Sensory Prosthesis, VR-based Rehabilitation, Gait Analysis, Ergonomics

Education

University of Illinois at Urbana-Champaign *Urbana, IL*
PH.D. MECHANICAL ENGINEERING *Aug 2006 - Dec 2010*

- Fields of Specialty: Controls, Dynamics and Applied Mathematics with Emphasis on Biomechanics and Postural Control
- Dissertation: Quantification of the human postural control system to perturbations
- Advisor: Elizabeth T. Hsiao-Wecksler
- Committee Members: Karl Rosengren, Srinivasa M. Salapaka, Prashant G. Mehta

M.S. APPLIED MATHEMATICS *Aug 2008 - May 2010*

- Fields of Specialty: Optimization and Algorithm
- Advisor: Karen Mortensen

Korea Advanced Institute of Science and Technology *Daejeon, Korea*
M.S. MECHANICAL ENGINEERING *Sep 2004 - Aug 2006*

- Fields of Specialty: Virtual Reality
- Thesis: HLA-based integration of underwater vehicle simulations using X3D multi-channel visualization and a motion platform
- Advisor: Soonhung Han
- Committee Members: Dong Soo Kown, Jung Kim

Hanyang University *Seoul, Korea*

B.S. Mechanical Engineering *Mar 1998 - Aug 2004*

- Fields of Interests: Robotics, Automatic Control, Mechatronics
- Thesis: Implementation of a clock using LEDs and inverted pendulum
- Advisor: Jahng-Hyon Park
- Summa Cum Laude

Busan High School *Busan, Korea*

High School Diploma *Mar 1995 - Feb 1998*

- Concentration on Mathematics and Science
- Summa Cum Laude

Publication

Journal Papers (Under Review and In Prep)

r8. Jiyoung Jeong, and **Pilwon Hur**, "Terrain-Adaptive Locomotion Prediction and Joint Trajectory Generation for Smooth Autonomous Transitions in Transfemoral Prostheses Using Estimated Stride Length," *IEEE RA-L*, Under review

r7. Hyungseok Ryu, and **Pilwon Hur**, "Active Inference-Based Implementation of Disturbance Observers: An Empirical Validation for Robotic Control," *IEEE RA-L*, In Preparation

r6. Sunwoong Moon, and **Pilwon Hur**, "Gait Pattern Adaptive Phase Variable Estimation Based on Adaptive Dynamic Movement Primitives," *IEEE RA-L*, In Preparation

r5. Hyewon Park, and **Pilwon Hur**, "Bi-LSTM based Estimation of Time-Varying Joint Impedance," *IEEE RA-L*, In Preparation

r4. Sunghwan Bae, and **Pilwon Hur**, "Addressing Non-Sparsity Issues in Non-Negative Matrix Factorization for Muscle Synergy Analysis," *IEEE TNSRE*, In Preparation

r3. Kang-Woo Lee, Yuyoung Kim, and **Pilwon Hur**, "Development and Interpretation of a Diagnostic Model for Foot and Ankle Diseases Using Public Medical Imaging," *Medical Image Analysis*, In Preparation

r2. Yi-Tsen Pan, and **Pilwon Hur**, "Type of skin stretch feedback affects postural sway differently: differential information may be more effective for balance rehabilitation," *Journal of NeuroEngineering and Rehabilitation*, Under review

r1. Mohammad Moein Nazifi, Isador H. Lieberman, Ram Haddas, and **Pilwon Hur**, "A muscle synergy approach in evaluation of the gait complexity following surgical alignment," *IEEE Trans on Neural Systems and Rehab Eng*, Under review

Journal Papers (Published)

j43. Kwonseung Cho, Kang-woo Lee, and **Pilwon Hur**, "Optimizing Toe Joint Stiffness to Improve Human-Like Walking," *Scientfic Reports*, Accepted

j42. Myeongju Cha, and **Pilwon Hur**, "Estimation of Gait Phase of Human Stair Descent Walking Based on Phase Variable Approach," *IEEE RA-L*, Vol 10, Issue 8, 2025

j41. Veronica Knisley, Kwonseung Cho, and **Pilwon Hur**, "Preferred Jacobian Differentiation and Direct Collocation Methods for an Efficient and Accurate Bipedal Walking Gait," *International Journal of Control, Automation, and Systems*, Vol 23, Issue 5, pp1551-1562, 2025

j40. Leonardo A. Elias, **Pilwon Hur**, and Gary C. Sieck, "Editorial: Towards an Understanding of Spinal and Corticospinal Pathways and Mechanisms," *Frontiers in Neurorobotics*, Vol 17, 2023

j39. Woolim Hong, Namita Anil Kumar, Shawanee Patrick, Sunwoong Moon, and **Pilwon Hur**, "A Feasibility Study of Piecewise Phase Variable Based on Variable Toe-Off for the Powered Prosthesis Control: A Case Study," *IEEE Robotics and Automation Letters*, Vol 8, Issue 5, pp2590-2597, 2023

j38. Woolim Hong, Jinwon Lee, and **Pilwon Hur**, "Piecewise Linear Labeling Method for Speed-Adaptability Enhancement in Human Gait Phase Estimation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, Vol 31, pp628-635, 2023

j37. Woolim Hong, Namita Anil Kumar, Shawanee Patrick, Hui-Jin Um, Heon-Su Kim, Hak-Sung Kim, and **Pilwon Hur**, "Empirical Validation of an Auxetic Structured Foot With the Powered Transfemoral Prosthesis," *IEEE Robotics and Automation Letters*, Vol 7 pp11228-11235, 2022

j36. Shawanee' Patrick, Namita Anil Kumar, and **Pilwon Hur**, "Evaluating Knee Mechanisms for Assistive Devices," *Frontiers in Neurorobotics*, Vol 16.790070, 2022 j35. Shawanee' Patrick, Namita Anil Kumar, and **Pilwon Hur**, "Biomechanical Impacts of Toe Joint with Transfemoral Amputee using a Powered Knee-Ankle Prosthesis," *Frontiers in Neurorobotics*, Vol 16.809380, 2022

j34. Woolim Hong, Jinwon Lee, and **Pilwon Hur**, "Effect of Torso Kinematics on Gait Phase Estimation for Enhancing Speed Adaptability," *Frontiers in Neurorobotics*, Vol 16.807826, 2022

j33. Namita Anil Kumar, and **Pilwon Hur**, "A Handheld Gyroscopic Device for Haptics and Hand Rehabilitation," *IEEE Transactions on Haptics*, Vol 15, pp109-114, 2022

j32. Namita Anil Kumar, Shawanee' Patrick, Woolim Hong, and **Pilwon Hur**, "Control Framework for Sloped Walking with a Powered Transfemoral Prosthesis," *Frontiers in Neurorobotics*, Vol 15:790060, 2022

j31. Hui-Jin Um, Heon-Su Kim, Woolim Hong, Hak-Sung Kim, and **Pilwon Hur**, "Design of 3D Printable Prosthetic Foot to Implement Nonlinear Stiffness Behavior of Human Toe Joint Based On Finite Element Analysis," *Scientific Reports*, Vol 11:19780, 2021

j30. Priscilla Lightsey, Yonghee Lee, Nancy Krenek, and **Pilwon Hur**, "Physical therapy treatments incorporating equine movement: a pilot study exploring interactions between children with cerebral palsy and the horse," *Journal of NeuroEngineering and Rehabilitation*, Vol 18:132, 2021

j29. Jinwon Lee, Woolim Hong, and **Pilwon Hur**, "Continuous Gait Phase Estimation using LSTM for Robotic Transfemoral Prosthesis Across Walking Speeds," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, Vol 29, pp1470-1477, 2021

j28. Woolim Hong, Namita Anil Kumar, and **Pilwon Hur**, "A phase-shifting based human gait phase estimation for powered transfemoral prostheses," *IEEE Robotics and Automation Letters*, Vol 6, Issue 3, pp5113-5120, 2021

j27. Mohammad Moein Nazifi, Kurt Beschorner, and **Pilwon Hur**, "Angular momentum regulation may dictate the slip severity," *PLOS One*, 15(3): e0230019, 2020

j26. **Pilwon Hur**, Yi-Tsen Pan, and Christian DeBuys, "Free energy principle in human postural control system: skin stretch feedback reduces the entropy," *Scientific Reports*, Vol 9:16870, 2019

j25. Mohammad Moein Nazifi, Kurt Beschorner, and **Pilwon Hur**, "Do walking muscle synergies influence propensity of severe slipping?," *Frontiers in Human Neuroscience*, Vol 13:383, 2019

j24. Na Jin Seo, Vincent Crocher, Egli Spaho, Charles Ewert, Mojtaba F. Fathi, **Pilwon Hur**, Suzanne Marchant, Michelle Woodbury, Sara J. Atkinson, Elizabeth M. Humanitzki, Abigail Lauer, "Capturing Upper Limb Gross Motor Categories Using the Kinect Sensor," *American Journal of Occupational Therapy*, Vol 73, Issue 3, 2019

j23. Mohammad Moein Nazifi, **Pilwon Hur**, Theodore Belanger, Akwasi Boah, Ram Haddas, Isador Lieberman, " The effect of surgical alignment on gait complexity in adult deformity patients: a neuromuscular synergy approach," *The Spine Journal*, Vol 18, Issue 8, S130, 2018

j22. **Pilwon Hur**, Yi-tsen Pan, Theodore Belanger, Isador Lieberman, Rajesh Arakal, Ram Haddas, " The effect of surgical alignment on standing balance in adult deformity patients: invariant density approach," *The Spine Journal*, Vol 18, Issue 8, S82, 2018

j21. Mohammad Moein Nazifi, Kurt Beschorner and **Pilwon Hur**, "Association between slip severity and muscle synergies of slipping," *Frontiers in Human Neuroscience*, Vol 11:536, 2017

j20. Han Ul Yoon, Namita Anil Kumar and **Pilwon Hur**, "Synergistic Effects on the Elderly People's Motor Control by Wearable Skin-Stretch Device Combined with Haptic Joystick," *Frontiers in Neurorobotics*, Vol 11:31, 2017

j19. Han Ul Yoon, Ranxiao F Wang, Seth A Hutchinson, and **Pilwon Hur**, "Customizing Haptic and Visual Feedback for Assistive Human-Robot Interface and the Effects on Performance Improvement," *Robotics and Autonomous Systems*, Vol 91, pp258-269, 2017

j18. Mohammad Moein Nazifi, Han Ul Yoon, Kurt Beschorner, and **Pilwon Hur**, "Shared and Task-Specific Muscle Synergies During Normal Walking and Slipping," *Frontiers in Human Neuroscience*, Vol 11:40, 2017

j17. Yi-Tsen Pan, Han U. Yoon, and **Pilwon Hur**, "A Portable Sensory Augmentation Device for Balance Rehabilitation Using Fingertip Skin Stretch Feedback," *IEEE Trans on Neural Systems and Rehab Eng*, Vol 25, Issue 1, pp28-36, 2017

j16. Na Jin Seo, Mojitaba Fathi-Firoozabad, **Pilwon Hur**, and Vincent Crocher, "Modifying Kinect Placement to Improve Upper Limb Joint Angle Measurement Accuracy," *Journal of Hand Therapy*, Vol 29, Issue 4, pp465-473, 2016

j15. Na Jin Seo, Jayashree Arun Kumar, **Pilwon Hur**, Vincent Crocher, Binal Motawar, Kishor Lakshminarayanan, "Development and usability evaluation of low-cost hand and arm virtual reality rehabilitation games," *Journal of Rehab Research and Development*, Vol 53, Issue 3, pp1-13, 2016

j14. **Pilwon Hur**, Kiwon Park, Karl Rosengren, Gavin Horn and Elizabeth Hsiao-Wecksle, "Effects of air bottle design on postural control of firefighters," *Applied Ergonomics*, Vol 48, pp49-55, 2015

j13. Na Jin Seo, Marcella Kosmopoulos, Leah Enders, **Pilwon Hur**, "Effect of Remote Sensory Noise on Hand Function Post Stroke," *Frontiers in Human Neuroscience*, Vol 8:934, 2014

j12. **Pilwon Hur**, Yao-Hung Wan, and Na Jin Seo, "Investigating the Role of Vibrotactile Noise in Early Response to Perturbation," *IEEE Trans on Biomed Eng*, Vol 61, Issue 6, pp1628-1633, 2014

j11. **Pilwon Hur**, Binal Motawar, and Na Jin Seo, "Muscular responses to handle perturbation with different glove condition," *Journal of Electromyography & Kinesiology*, Vol 24, Issue 1, pp159-164, 2014

j10. **Pilwon Hur**, Karl Rosengren, Gavin Horn, Denise Smith and Elizabeth Hsiao-Wecksler, "Effect of protective clothing and fatigue on functional balance of firefighters," *J Ergonomics*, S2:004, 2013

j9. Leah R. Enders, **Pilwon Hur**, Michelle J. Johnson, and Na Jin Seo, "Remote vibrotactile noise improves light touch sensation in stroke survivors' fingertips via stochastic resonance," *Journal of NeuroEng and Rehab*, 10:105, 2013

j8. **Pilwon Hur**, Binal Motawar, and Na Jin Seo, "Hand breakaway strength model −Effects of glove use and handle shapes on a person's hand strength to hold onto handles to prevent fall from elevation," *Journal of Biomechanics*, Vol 45, Issue 6, pp958-964, 2012

j7. Binal Motawar, **Pilwon Hur**, James Stinear, and Na Jin Seo, "Contribution of intracortical inhibition in voluntary muscle relaxation," *Exp Brain Research*, Vol 221, Issue 3, pp299-308, 2012

j6. **Pilwon Hur**, A. Kenneth Shorter, Prashant Mehta, and Elizabeth Hsiao-Wecksler, "Invariant Density Analysis: modeling and analysis of the postural control system using Markov chains," *IEEE Trans on Biomed Eng*, Vol 59, Issue 4, pp1094-1100, 2012

j5. Kiwon Park, **Pilwon Hur**, Karl Rosengren, Gavin Horn, and Elizabeth Hsiao-Wecksler, "Effect of load carriage on gait due to firefighting air bottle configuration," *Ergonomics*, Vol 53, Issue 7, pp882-891, 2010

j4. **Pilwon Hur**, Brett Duiser, Srinivasa Salapaka, and Elizabeth Hsiao-Wecksler, "Measuring robustness of the postural control system to a mild impulsive perturbation," *IEEE Trans on Neural Systems and Rehab Eng*, Vol 18, Issue 4, pp461-467, 2010

j3. Gavin Horn, Elizabeth Hsiao-Wecksler, Karl Rosengren, **Pilwon Hur**, Kiwon Park, and Denise Smith, "Slips, trips, and falls on the fireground - A study at IFSI," *Fire Rescue*, Vol 27, Issue 1, pp56-58, 2009

j2. **Pilwon Hur**, Byounghyun Yoo, Jeongsam Yang, and Soonhung Han, "An underwater vehicle simulator with immersive interface using X3D and HLA," *SIMULATION, Trans of the Society for Modeling and Simulation International*, Vol 85, Issue 1, pp33-44, 2009

j1. **Pilwon Hur**, Jeongsam Yang, and Soonhung Han, "An Underwater Vehicle Simulator using X3D and a Motion Chair in a Multi-Channel Display Room," *Soc CAD/CAM Eng*, Vol 13, Issue 1, pp45-57, 2008

Conference Papers and Presentations

c143. 주진식, 허민구, 허필원, "방사성 동위원소 취급을 위한 Master-Slave 매니퓰레이터 시스템 개발," *ICROS*, 2025년 6월 25-27일, 전주

c142. Suyeong Park, and **Pilwon Hur**, "LSTM-Based Real-Time Classification of Locomotion States Using Inertial Measurement Unit (IMU) Data," *ICROS*, 2025년 6월 25-27일, 전주

c141. Hyungseok Ryu, and **Pilwon Hur**, "Active Inference-Based Control and Disturbance Torque Estimation," *ICROS*, 2025년 6월 25-27일, 전주

c140. Kwonseung Cho, and **Pilwon Hur**, "Development of a Lower Limb Exoskeleton for Gait Rehabilitation in Hemiparetic Patients," *ICROS*, 2025년 6월 25-27일, 전주

c139. Jiyoon Sung, and **Pilwon Hur**, "FPCA-Based Balance Recovery Control Strategies Under Perturbation Conditions," *ICROS*, 2025년 6월 25-27일, 전주

c138. Laeheon Kim, and **Pilwon Hur**, "Double-Inverted Pendulum Control With One Input Based on MDP," *ICROS*, 2025년 6월 25-27일, 전주

c137. Sunghwan Bae, and **Pilwon Hur**, "Improving the Non-Sparsity problem of Non-negative Matrix Factoriztion for muscle synergy-based Gait Intention Estimation," *ICROS*, 2025년 6월 25-27일, 전주

c136. Myeongju Cha, Hyeon Cho, Laeheon Kim, and **Pilwon Hur**, "Design and Trajectory Tracking Control of a Customizable Active Prosthesis Leg," *ICROS*, 2025년 6월 25-27일, 전주

c135. Sunwoong Moon, and **Pilwon Hur**, "Phase Variable Estimation Based on Gait Pattern Adaptive Dynamic Movement Primitives," *ICROS*, 2025년 6월 25-27일, 전주

c134. Sunghyun Kim, and **Pilwon Hur**, "Design Proposal of a Powered Knee Exoskeleton Considering Wearability," *ICROS*, 2025년 6월 25-27일, 전주

c133. Hyewon Park, and **Pilwon Hur**, "Bi-LSTM based Estimation of Time-Varying Joint Impedance: Comparison with Classical Methods Using Simulation," *ICROS*, 2025년 6월 25-27일, 전주

c132. Kwonseung Cho, Kang-Woo Lee, and **Pilwon Hur**, "Optimizing Toe Joint Stiffness to Improve Human-Like Walking," 대한기계학회 바이오공학부문 춘계학술대회, 2025년 5월 21-23일, 부산

c131. Hyeon Cho, and **Pilwon Hur**, "Effects of Walking Speed and Visual Stimulus on the Axis Orientation of the Metatarsophalangeal Joint and Foot Progression Angle During Gait," 대한기계학회 바이오공학부문 춘계학술대회, 2025년 5월 21-23일, 부산

c130. Kang-Woo Lee, and **Pilwon Hur**, "Localized Morphological Differentiation of Foot Disorders via Statistical Shape Modeling," 대한기계학회 바이오공학부문 춘계학술대회, 2025년 5월 21-23일, 부산

c129. Jinsik Ju, Young Bae Kong, Jongchul Lee, Jeonghoon Park, **Pilwon Hur**, and Mingoo Hur, "Development of a remote control system for a syringe pump using a dynamixel motor," *ICCAS*, Oct 29-Nov 1, 2024, Jeju, Korea

c128. Jiyoung Jeong, and **Pilwon Hur**, "Environment perception and joint trajectory generation for robotic transfemoral prosthesis," 대한기계학회 바이오공학부문 춘계학술대회, 2024년 4월 24-26일, 여수

c127. Hyeon Cho, and **Pilwon Hur**, "Analysis of influence of the conscious and subconscious walking on the orientation of human toe joint axis at various walking speeds," 대한기계학회 바이오공학부문 춘계학술대회, 2024년 4월 24-26일, 여수

c126. Wooyoung Park, and **Pilwon Hur**, "Data-driven control of gait-assistive wearable robots using GPR," *ICROS*, 2024년 7월 2-4일, 대전

c125. 안해원, 허필원, "데이터 기반 모델링을 이용한 2-Link 인간 모델의 균형 제어," *ICROS*, 2024년 7월 2-4일, 대전

c124. Sunghwan Bae, and **Pilwon Hur**, "Improving Nonnegative Matrix Factorization for Muscle Synergy Analysis: Minimizing Synergy Entropy," *ICROS*, 2024년 7월 2-4일, 대전

c123. Sunwoong Moon, Myeongju Cha, Hyungseok Ryu, Kwonseung Cho, Jiyoon Sung, Kangwoo Lee, Sunghyun Kim, and **Pilwon Hur**, "State Estimation of Nonlinear System Based on Free Energy Principle," *ICROS*, 2024년 7월 2-4일, 대전

c122. 박혜원, 허필원, "휴대용 CPM(Continuous Passive Motion) 기구 제어를 위한 딥러닝 기반 사람 관절 stiffness 추정," *ICROS*, 2024년 7월 2-4일, 대전

c121. 정지영, 허필원, "착용형 보행보조로봇을 위한 환경인지 및 관절 경로 생성," *ICROS*, 2024년 7월 2-4일, 대전

c120. 차명주, 허필원, "계단 하강 보행의 상태 예측을 위한 상태 변수 생성에 대한 연구," *ICROS*, 2024년 7월 2-4일, 대전

c119. Woolim Hong, Namita Anil Kumar, Shawanee Patrick, Sunwoong Moon, and **Pilwon Hur**, "A Feasibility Study of Piecewise Phase Variable Based on Variable Toe-Off for the Powered Prosthesis Control: A Case Study," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct 1-5, Detroit, USA, 2023

c118. HyungSeok Ryu, Woolim Hong, and **Pilwon Hur**, "Towards Realistic Prosthetic Gait Simulations: Enhancing the Accuracy of OpenSim Analysis by Integrating the Transfemoral Prosthesis Model," *IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, Aug 28-31, Busan, Korea, 2023

c117. Jinsik Ju, Junyoung Lee, Jeonghoon Park, **Pilwon Hur**, and Mingoo Hur, "Development of an Automated Program for the Radioisotope Separation and Purification System," *ICCAS*, Oct 17-20, 2023, Yeosu, Korea

c116. 류형석, 홍우림, 허필원, "현실적인 의족 보행 시뮬레이션을 향해: 대퇴의족 모델을 통합하여 OpenSim 분석의 정확도 향상하기," 대한기계학회 바이오공학부문 춘계학술대회, 2023년 4월 27일, 강릉

c115. Veronica Knisley, Kwonseung Cho, and **Pilwon Hur**, "Preferred Jacobian Differentiation and Direct Collocation Method for an Efficient and Accurate Bipedal Walking Gait," *ICROS*, 2023년 6월 21일, 삼척

c114. 류형석, 허필원, "환측의 IMU와 보행시점 정보를 활용한 보행 비대칭성과 보행 속도 추정," 대한생체역학회, 2023년 12월 1일, 전주

c113. 문선웅, 허필원, "aDMP based Human Gait Pattern adaptive Phase Variable Estimation," 대한생체역학회, 2023년 12월 1일, 전주

c112. 차명주, 허필원, "계단 하강 보행 상태 추정을 위한 상태 변수 생성," 대한생체역학회, 2023년 12월 1일, 전주

c111. 성지윤, 허필원, "자유 에너지의 전반적인 원리와 이해," 대한생체역학회, 2023년 12월 2일, 전주

c110. 조권승, 허필원, "Effect of Toe Joint on Bipedal Walking," 대한생체역학회, 2023년 12월 2일, 전주

c109. Woolim Hong, Namita Anil Kumar, Shawanee Patrick, Hui-Jin Um, Heon-Su Kim, Hak-Sung Kim, and **Pilwon Hur**, "Empirical Validation of an Auxetic Structured Foot With the Powered Transfemoral Prosthesis," *IEEE RAS/EMBS International Conference on Biomedical Robotics & Biomechatronics*, Aug 21 - 24, Seoul, Korea, 2022

c108. 홍우림, 이진원, 허필원, "다양한 속도에서의 하지 웨어러블 로봇 제어를 위한 기계학습 기반 사용자 보행 위상 추정," 대한기계학회 *IT*융합부분 춘계학술대회, 2022년 5월25일-27일, 안동

c107. 조현, 엄희진, 김헌수, 홍우림, 김학성, 허필원, "FEA를 통한 인간발가락 관절의 비선형 강성거동을 구현하기 위한 3D 프린팅이 가능한 의족 발," 대한기계학회 *IT*융합부분 춘계학술대회, 2022년 5월25일-27일, 안동

c106. 정지영, 이진원, 홍우림, 허필원, "로봇 의족을 위한 LSTM을 이용한 여러 보행속도에서의 연속 보행단계 추정," 대한기계학회 *IT*융합부분 춘계학술대회, 2022년 5월25일-27일, 안동

c105. 류형석, 홍우림, 이진원, 허필원, "보행속도에 따른 보행위상추정에 몸통 운동학의 영향," 대한기계학회 *IT*융합부분 춘계학술대회, 2022년 5월25일-27일, 안동

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학적 영향," 대한기계학회 *IT*융합부분 춘계학술대회, 2022년 5월25일-27일, 안동

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b2. **Pilwon Hur**, Kenneth Chao, and Victor Christian Paredes Cauna, "Unification of bipedal robotic walking using quadratic program-based control Lyapunov function: applications to regulation of ZMP and angular momentum" In Dan Zhang (Ed.) *Adaptive Control for Robotic Manipulators* CRC Press/Taylor & Francis Group, Boca Raton, FL, Nov 2016

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i76. "자유 에너지 원리에 기반한 착용형 보행 보조 로봇 제어," 대한기계학회 바이오공학부문 하계 워크샵, KAIST, 2025년 8월 21일

i75. "감각증강을 통한 보행 및 균형 향상, 그리고 보조로봇에 대한 적용 가능성," 재활로봇 측정 및 평가 워크샵, 국립재활원, 2024년 9월 24일

i74. "Linking neuromechanics and robotics in human movement," *Rehabilitation Research Seminar*, National Rehabilitation Center,, Korea, June 13, 2024

i73. "The mechanisms of self-organizing systems with multi-scale embodied intelligence," *Multi-agent workshop*, GIST, Korea, April 2, 2024

i72. "The mechanisms of self-organizing systems with multi-scale embodied intelligence," *Rehabilitation Robotics Workshop in Korea Robotics Conference*, Pyeongchang, Korea, Feb 21, 2024

i71. "Towards an Understanding of Spinal and Corticospinal Pathways and Mechanisms," *BCI & Neurorehabilitation Workshop*, National Rehabilitation Center, Seoul, Korea, Nov 9, 2023

i70. "Free Energy Principle: A Unifying Framework for Ation, Perception, and Learning of Dynamic Systems Involving Robots and/or Human Inspired by Neuroscience," *Department of Smart Industrial Machine Technologies Seminar Series*, Korea Institute of Machinery & Materials, Daejon, Korea, Oct 25, 2023

i69. "Free Energy Principle: A Unifying Framework for Ation, Perception, and Learning of Dynamic Systems Involving Robots and/or Human Inspired by Neuroscience," *GIST Math*, GIST, Korea, Sep 26, 2023

i68. "Neuromechanics in Human Robot Interaction," *LG Electronics Webinar*, Korea, Sep 26, 2023

i67. "Human Robot Interaction," *Gwangju Science Museum*, Gwangju, Korea, Jul 19, 2023

i66. "Human Centric Framework for the Physical Human Robot Interaction," *Suncheon Jeil High School Lecture Series*, Suncheon, Korea, Jul 5, 2023

i65. "Understanding Linear Control Systems from the Operator-Theoretic Perspectives," *ME Seminar*, GIST, Korea, Jun 8, 2023

i64. "Free Energy Principle in Postural Control: Analysis and Rehabilitation," *Rehabilitation Research Seminar*, National Rehabilitation Center, Seoul, Korea, Dec 14, 2022

i63. "Introduction to Mathematica for the Mathematics Education in High School," *GSA Seminar*, Gwangju Science Academy, Gwangju, Korea, Nov 16, 2022

i62. "Human Centric Framework for the Physical Human Robot Interaction," *Korean Minjok Leadership Academy Lecture*, GIST, Korea, Oct 26, 2022 i61. "Bipedal Walking: Human's View vs. Robot's View" *Colloquium in Mechanical Engineering*, Seoul National University, Korea, Oct 7, 2022

i60. "Human Centric Framework for the Physical Human Robot Interaction," *KiTech Seminar*, KiTech, Gwangju, Korea, Sep 20, 2022

i59. "Bipedal Walking: Human's View vs. Robot's View" *ICROS Control Theory Workshop*, Inje, Korea, Sep 2, 2022

i58. "Human Centric Framework for the Physical Human Robot Interaction," *Hyundai Motor Company Seminar*, GIST, Korea, Aug 29, 2022

i57. "Human Centric Framework for the Physical Human Robot Interaction," *Slovakia Collaboration Seminar*, GIST, Korea, Jun 9, 2022

i56. "Human Rehabilitation Group: From *F=ma* to Complex Robots and Biomechanics" *Thematic Lecture: Mechanical Engineering*, Incheon POSCO High School, Incheon, May 27, 2022

i55. "Human Centric Framework for the Physical Human Robot Interaction," *GIST ME Seminar*, GIST, Korea, Nov 26, 2021

i54. "Physical therapy treatments incorporating equine movement: a pilot study exploring interactions between children with cerebral palsy and the horse," *Korean Association of Therapeutic Horsemanship*, Samsung Medical Center, Korea, Nov 20, 2021

i53. "Human Centric Framework for the Physical Human Robot Interaction," *GIST-LG Conference*, GIST, Korea, Nov 19, 2021

i52. "Linking Neuromechanics and Robotics in Human Movement," *Colloquium in Mechanical Engineering*, Seoul National University, Korea, April 16, 2021

i51. "Linking Neuromechanics and Robotics in Bipedal Walking," *Korean Society of Sport Biomechanics*, Hallym University, Korea, Dec 18 - 19, 2020

i50. "Motor Control and Rehabilitation of Human Movement," *Kinesiology Seminar*, Seoul National University, Korea, June 24, 2020

i49. "Dynamics and Control of Robust Bipedal Walking," *ME Seminar*, GIST, Gwangju, Korea, June 16, 2020

i48. "Motor Control and Rehabilitation of Human Movement," *Robotics Seminar*, DGIST, Daegue, Korea, June 12, 2020

i47. "Academic Careers in the United States," *Colloquium in Mechanical Engineering*, Seoul National University, Korea, May 15, 2020

i46. "Human Robot Interaction for Rehabilitation," *NSF Workshop on Human-Friendly Robots*, University of Texas San Antonio, San Antonio, TX, May 2-3, 2019

i45. "Human Robot Interaction and Free Energy Principle," *Technology Collaboration Center*, NASA JCS, Houston, TX, March 28, 2019

i44. "Links between Robotics and Biomechanics for Gait and Balance Rehabilitation," *TAMU Kinesiology Seminar Series*, College Station, TX, Nov 16, 2018

i43. [Key Note]"Links between Robotics and Biomechanics for Gait and Balance Rehabilitation," *Southwest Texas Asian Symposium*, Corpus Christi, TX, Nov 2, 2018

i42. "Young Generation Forum: How to survive in academia," *KSEA*, Houston, Aug 25, 2018

i41. "Effect of toe stiffness on the push-off and joint trajectories for the powered transfemoral prosthesis," *UKC 2018*, New York, Aug 3, 2018

i40. "Rehabilitation robotics for human locomotion," *Mechanical Engineering, University of Cape Town*, Cape Town, South Africa, Jun 27, 2018

i39. [Key Note]"Optimality in human balance and walking," *IX International Seminar of Biomedical Engineering, Uniandes*, Bogota, Colombia, May 17, 2018

i38. [Lecture with Certificate]"Modeling and control of human balance and walking," *IX International Seminar of Biomedical Engineering, Uniandes*, Bogota, Colombia, May 16, 2018

i37. "Enhancing biomechanics of prosthetic walking with the powered transfemoral prosthesis," *Orthotic & Prosthetic Innovative Technologies Conference*, University of California, San Francisco, May 12, 2018

i36. "Robotics Research in Human Rehabilitation: Gait and Balance," *Mechanical Engineering Seminar, University of Texas at San Antonio*, San Antonio, TX, Mar 30, 2018

i35. "Robotics Research in Human Rehabilitation: Gait and Balance," *College of Engineering Seminar, TAMU Corpus Christi*, Corpus Christi, TX, Mar 29, 2018

i34. "Robotics Research in Human Rehabilitation: Gait and Balance," *Informational for TURTLE Meeting, TAMU*, College Station, TX, Mar 21, 2018

i33. "Mechanical Engineering for Rehabilitation Robotics," *MEEN 289 Seminar, TAMU*, College Station, TX, Nov 10, 2017 i32. “Optimality and Robustness of Human Walking and Balance,” *Department of Biomedical Engineering*, University of Los Andes, Bogota, Colombia, Jun 12, 2017

i31. “Enhancing Balance of the Aged Workers Via Sensory Augmentation Toward the Reduction of Injuries Due to Falls,” *NIOSH ERC Pilot Projects Research Symposium*, UTHealth School of Public Health, Houston, TX, Jun 2, 2017

i30. “Neuro/Biomechanics for Gait and Balance Rehabilitation,” *Korean Aggies Bio Association*, Texas A&M University, College Station, TX, Apr 28, 2017

i29. [Key Note]“Gait and Balance Rehabilitation,” *South Central ASB Regional Conference*, Texas Back Institute, Plano, TX, Apr 1, 2017

i28. “Robotics in Gait and Balance Rehabilitation,” *Health Science Seminar, University of Houston*, Houston, Feb 14, 2017

i27. “Neuro/biomechanics and its application in robotics,” *MEEN 681 Seminar, TAMU*, College Station, TX, Jan 25, 2017

i26. “Gait and Balance Rehabilitation,” *West Gulf Coast Regional Conference, Korean-American Scientists and Engineers Association*, Houston, Nov 19, 2016

i25. “Human-Like Walking and its Robustness to Perturbations,” *Robotics Engineering, DGIST*, Daegu, Korea, Oct 17, 2016

i24. “Linking Robotics and Neuro/Biomechanics,” *Mechanical Engineering, Hanyang University*, Seoul, Korea, Oct 14, 2016

i23. “Neuro/biomechanics and its applications in Rehabilitation Robotics,” *Mechanical Engineering, GIST*, Gwangju, Korea, Oct 13, 2016

i22. “Neuro/biomechanics and its applications in Robotics,” *Mechanical Engineering, KAIST*, Daejon, Korea, Oct 12, 2016

i21. “Muscle Synergy as a Rehabilitation Tool,” *The 1st Kinesiology Seminar, Seoul National University*, Seoul, Korea, Oct 10, 2016

i20. “Usage of MATLAB in rehabilitaiton robotics,” *Workshop on Scientific Computing with MATLAB at Texas A&M in conjunction with Dr. Cleve Moler’s Talk on Evolution fo MATLAB*, College Station, TX, Apr 25, 2016

i19. “From F=ma to complex robots and biomechanics,” *MEEN Informational hosted by Mechanical Engineering Leadership Council*, College Station, TX, Mar 31, 2016

i18. “Development of wearable balance rehabilitation device for the low-income elderly adults in Seoul,” Seoul City Forum, *US-Korea Conference*, Atlanta, GA, Jul 31, 2015

i17. “Collaboration with human rehabilitation group,” *Lynntech*, College Station, TX, May 6, 2015

i16. “Artificial sensory augmentation via skin stretch feedback and its effect in postural control,” TAMU ENG-LIFE 2015 Workshop, *TAMU*, College Station, TX, Apr 24, 2015

i15. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” Kinesiology Seminar, *TAMU*, College Station, TX, Feb 20, 2015

i14. “Upper and lower limbs rehabilitation for the elderly and neurologically impaired patients,” *Korea Institute of Industrial Technology*, Seongnam, Korea, Aug 14, 2014

i13. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *GIST*, Gwangju, Korea, Aug 13, 2014

i12. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *Korea Institute of Industrial Technology*, Ansan, Korea, Jul 30, 2014

i11. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *Kyunghee University*, Suwon, Korea, Jul 30, 2014

i10. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *KAIST*, Daejeon, Korea, Jul 22, 2014

i9. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *Ajou University*, Suwon, Korea, Jul 21, 2014

i8. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *University of Tennessee-Knoxville*, Knoxville, TN, Feb 17, 2014

i7. “Optimality of human movement: diagnosis and rehabilitation for enhanced quality of life,” *Texas A&M University*, College Station, TX, Feb 11, 2014

i6. “Improving quality of life: understanding fall mechanisms and potential fall preventions,” *Florida International University*, Miami, FL, Nov 6, 2013

i5. “Improving assistive gloves for stroke survivors using dynamic biomechanical models and optimization,” R24-Engineering for Neurological Rehabilitation Meeting, *Rehabilitation Institute of Chicago*, Chicago, IL, Jun 16, 2013

i4. “Upper and lower limbs rehabilitation for the neurologically impaired patients,” *DGIST*, Daegu, Korea, Feb 1, 2013

i3. “How mechanical noise enhance human sensation?,” Series of Seminars at the Center for u-Healthcare, *Soon Chun Hyang*

University, Asan, Korea, Oct 29, 2012

i2. "Rehabilitation of the patients with physical weakness and neurologic impairments," *Hanyang University*, Seoul, Korea, Oct 27, 2012

i1. "Sensory enhancement via vibrotactile stimulation and its effect on the motor response post stroke," *DGIST*, Daegu, Korea, Oct 22, 2012

Exhibition

e1. Hyojin Jang, **Pilwon Hur**, Jinsil Hwaryoung Seo, "Lotus Flower, the Public Art," *24th International Symposium on Electronic Art*, Durban, South Africa, Jun 23 - 30, 2018

Patent

- Daniel McGowan, and **Pilwon Hur**, "Light Weight, Modular, Powered Transfemoral Prosthesis." US11596530B2, March 7th, 2023
- Daniel McGowan, and **Pilwon Hur**, "Light Weight, Modular, Powered Transfemoral Prosthesis." International Patent (PCT), WO/2019/118534, filed Dec 12, 2018
- **Pilwon Hur**, and Namita Anil Kumar, "Portable Gyroscopic Devices and Methods." U.S. Patent 62/413,130, filed Oct 26, 2016

Research Experience

Gwangju Insitute of Science and Technology *Gwangju, Korea*

ASSOCIATE PROFESSOR *Dec 2020 - PRESENT*

- Development of an AI-based modular upper limb rehabilitation for hemiplegia
- Design and control of bipedal walking of a humanoid robot with both model-based control and reinforcement learning
- Development of a control framework that unifies the action, estimation, and learning via FEP
- Design and control of exoskeleton for the hemiparetic patients
- Design and control of robotic knee assistive device using FEP
- Human-robot interaction, recognition of intent and trust via FEP
- Trajectory optimization of the bipedal systems for the human-like walking on various terrains with uncertainties
- Robotic rehabilitation framework using myoelectric control interface with skin stretch feedback for the upper limb deficiency patient
- Design and control of robotic transfemoral prosthesis
- Control of bipedal robotic walking with slip/push recovery
- Balance rehabilitation using sensory augmentation via skin stretch feedbacks
- Muscle synergy analysis and rehabilitation for human movement
- Analysis and design of toe joint for natural human and robotic walking
- Real time estimation of human joint impedance
- Development of tools for motion detection and analysis of national archery athletes
- Development of automatic scoring systems for archery competitions

Texas A&M University *College Station, TX*

Assistant Professor *Aug 2014 - Dec 2020*

- Human-Robot Interaction, Recognition of Intent and Trust via Free Energy Principle
- Trajectory optimization of the bipedal systems for the human-like walking on various terrains with uncertainties
- Robotic rehabilitation framework using myoelectric control interface with skin stretch feedback for the upper limb deficiency patient
- Design and control of robotic transfemoral prosthesis
- Shared control of the exoskeleton with compliant human-robot interaction
- Control of bipedal robotic walking with slip/push recovery
- Upper limb rehabilitation for the stroke survivors using gyroscopic effect
- Balance rehabilitation using sensory augmentation via skin stretch and electrotactile feedbacks
- Identification of cognitive status of the cerebral palsy patients and its efficacy on the equine-assisted therapy
- Development of the sensors for real-time monitoring of the equine-assisted therapy
- Haptic customization for the novice wheelchair driver for optimal performance
- Free energy principle for neuroscience and its application to human gait and balance
- Muscle synergy-based analysis and rehabilitation for slipping
- Robotic surface grinding

University of Wisconsin-Milwaukee *Milwaukee, WI*

Postdoctoral Research Fellow *Sep 2010 - Jun 2014*

- Identification of mechanism of sensory enhancement due to vibrotactile stimulation using EEG
- Design and biomechanical analysis of an exoskeleton glove for the hand rehabilitation for stroke patients
- Somatosensation enhancement and its effect on motor performance post stroke
- Validation of Microsoft kinect for upperlimb rehabilitation
- Development and usability test for virtual rehabilitation games for stroke patients
- Development of devices for quantifying biceps spasticity for stroke survivors
- Investigation of Botulinum toxin of the long finger flexor muscles on grip force control and muscle activation post stroke
- Design of a unilateral repetitive motion device (URMD) for the hand rehabilitation for stroke patients
- Effect of vibrotactile stimulation on tactile sensation post stroke
- Investigation of the role of sensory systems in detecting slip and fall accidents
- Investigation of the effect of cutaneous sensory enhancement on the reaction time to perturbation
- Optimal handle design and the breakaway strength to prevent falls from the elevation
- Investigation of the contribution of intracortical inhibition during voluntary muscle contraction
- Investigation of the effect of the handle shapes and coefficient of friction on breakaway strength
- Investigation of the effect of cutaneous sensation and friction coefficients on reaction time to handle perturbation

University of Illinois at Urbana-Champaign *Urbana, IL*

Research Assistant *Aug 2006 - Dec 2010*

- Development of a fall risk prediction model of the elderly
- Development of a novel stochastic method for analyzing center of pressure movement using Markov chains
- Investigation of the effect of air bottle configuration on the balance and gait performance of the firefighters
- Design of foot clearance sensing device.
- Investigation of the effect of fatigue on the balance of the firefighters
- Development of an assessment tool for functional balance of the firefighters after strenuous activities
- Development of a novel method for robustness quantification of the human postural control system to an external perturbation

Korean Advanced Institute of Science and Technology *Daejeon, Korea*

Research Assistant *Sep 2004 - Aug 2006*

- Integration of underwater vehicle simulators with multi-channel display system and motion platform over HLA (High Level Architecture)
- Integration of virtual environments for national science museum

Research Grants

g21. Pilwon Hur (PI) "Translational Research on Modular Upper Limb Rehabilitation Robot for Improving Upper Limb Function in Patients with Brain Lesions," Ministry of Health and Welfare, May 2025 - Nov 2027

g20. Pilwon Hur (PI) "AI-Based Human-Robot Collaboration in Extreme Environments via Humanoid Robots," GIST, Jan 2025 - Dec 2026

g19. Pilwon Hur (PI) "Preliminary research on obstacle avoidance and path planning of robot cleaners," LG electronics, May 2024 - Jul 2024

g18. Pilwon Hur (Co-PI) "AI-based Platform for Motion Analysis and Data Integration for Improving National Archery Performance," Ministry of Culture, Sports and Tourism, Jun 2023 - May 2025

g17. Pilwon Hur (Co-PI) "Development of AI-based Exoskeleton Robots for the Lower Limb Rehabilitation for the Hemiplegic Patients," Ministry of Health and Welfare, April 2023 - Nov 2025

g16. Pilwon Hur (PI) "Enhancing Performance of Wearable Lower Limb Assistive Robots using Sensorimotor Control Theory," Ministry of Health and Welfare, April 2023 - Nov 2025

g15. Pilwon Hur (PI) "Development of smart rehabilitation robots to resolve joint rigidity," Sunhan Hospital, Nov 2022 - Oct 2024

g14. Pilwon Hur (PI) "Development of human-robot interaction systems for lowerlimb amputee patients," GIST, Jan 2021 - Dec 2022

g13. Pilwon Hur (PI), James Hubbard (Co-PI) "Framework Development of Equine-Assisted Therapy for Children with Cerebral Palsy," MEEN Research Priming Seed Grant, April 2019 - April 2020

g12. Pilwon Hur (Co-PI), "Robotic Surface Grinding," Advanced Robotics Manufacturing (ARM) Institute, Sep 2018 - Nov 2019

g11. Deanna Kennedy (PI), Pilwon Hur (Co-PI), Li Zeng (Co-PI) "Integrated Feedback And Augmented Reality For Individuals With Motor Impairments," Texas A&M Triads for Transformation (T3), Jan 2019 - Dec 2020

g10. Pilwon Hur (PI), "Biomechanical Consideration of the Lower Limb Mobility Assistive Devices," National Science Foundation (NSF) AGEP, Sep 2018 - Jan 2019

g9. Pilwon Hur (PI), "A Low-Cost Motion Capture System using Smartphones to Resolve Healthcare Issues in Low-Income Countries," Aggie Challenge, Sep 2018 - May 2020

g8. Pilwon Hur (PI), Duane Steward, Nancy Krenek, Priscilla Lightsey, "Tracking kinematic and kinetic data during horse riding for optimizing therapeutic outcomes," Horses and Humans Research Foundation (HHRF), Jan 2018 - Jun 2019

g7. Pilwon Hur (PI), Reza Langari (Co-PI), Byung-Jun Yoon (Co-PI) "Robotic rehabilitation framework for the upper limb deficiency patient via myoelectric control interface with skin stretch feedback," PESCA, Division of Research, TAMU, May. 2017 - Apr. 2018

g6. Pilwon Hur (PI), "Enhancing balance of the aged workers via sensory augmentation toward the reduction of injuries due to falls," NIOSH, Pilot Project Research Training, 5T42OH008421, Jul. 2016 - Jun. 2017

g5. Pilwon Hur (PI), "Effect of enhancement of somatosensation on hand function post stroke," American Heart Association (AHA), Postdoc Fellowship, 12POST12090039, Jul. 2012 - Jun. 2014

g4. Na Jin Seo (PI), Pilwon Hur (Co-I), "Improving assistive gloves for stroke survivors using dynamic biomechanical models and optimization," NIH R24- Rehabilitation Engineering Research Network Center, Jul. 2012 - Jun. 2013

g3. Pilwon Hur (PI), "Development of a biomechanical hand model to predict multi-segmental finger flexion forces," NIOSH, Pilot Project Research Training, T42-OH008672, Jul. 2012 - Jun. 2013

g2. Pilwon Hur (PI), "Prevention of ladder fall by improved somatosensation and optimal rung design," National Institute for Occupational Safety and Health (NIOSH), Pilot Project Research Training, T42-OH008672, Jul. 2011 - Jun. 2012

g1. Tchekanov (PI), Na Jin Seo (Co-PI), Pilwon Hur (Co-I), "Effect of botulinum toxin of the long finger flexor muscles on grip force control following stroke," Physical Medicine and Rehabilitation at Medical College of Wisconsin, Jan. 2011 - Dec. 2013

Teaching Experience

Gwangju Institute of Science and Technology *Gwangju, Korea*

Associate Professor *Dec 2020 - PRESENT*

- t6. MC2103: Dynamics - FA2021, FA2022, FA2023, FA2024, FA2025
- t5. MC3215: Mechanical Signals and Systems - SP2021, SP2022, SP2023, SP2024, SP2025
- t4. ME5160: Advanced Dynamics - SP2023
- t3. ME5166: Intelligent Motor Integration Project - FA2023, FA2024, FA2025
- t2. ME5169: Mathematical Foundations of Optimization in Robotics - SP2025
- t1. ME8102: Multivariable Robust Control - SP2024

Texas A&M University *College Station, TX*

Assistant Professor *Aug 2014 - Dec 2020*

- t8. MEEN357: Engineering Analysis for Mechanical Engineers - SU2019, SU2020
- t7. MEEN364: Dynamic Systems and Controls - SP2015
- t6. MEEN404: Engineering Laboratory - FA2018
- t5. MEEN408/612: Introduction to Robotics/Mechanics of Robotic Manipulators - FA2014, FA2015, FA2016, FA2017, SP2020
- t4. MEEN431: Advanced System Dynamics and Controls - FA2016, FA2017, FA2019
- t3. MEEN651: Control System Design - FA2018, FA2019
- t2. MEEN652: Multivariable Control System Design - SP2017, SP2018, SP2019
- t1. MEEN655: Design of Nonlinear Control Systems - SP2016

Short Lectures, Crash Courses -

As a Faculty Position *Aug 2014 - PRESENT*

- t15. Winter Motor Boot Camp for GIST students, 2/24/2025-2/28/2025
- t14. Deep Reinforcement Learning (DRL) for GIST students, 8/21/2024
- t13. Markov Decision Process (MDP) for GIST students, 8/5/2024
- t12. Summer Motor Boot Camp for GIST students, 6/25/2024-6/28/2024
- t11. Understanding Linear Control Systems from Operator-Theoretic Perspectives for GIST graduate students, 6/8/2023
- t10. Perspectives on Optimal Control Problems for GIST Systems, Controls, and Robotics (SCR) Workshop, 2/21/2023
- t9. Trajectory optimization for bipedal walking using direct collocation, Tutorial for MEEN grad students, 1/17/2020,1/24/2020,1/31/2020
- t8. Unified Approach for Optimal Control using Functional Analysis, Tutorial for MEEN grad students, Dec, FA2019
- t7. Optimal Control (Calculus of Variation, Dynamic Programming, Pontryagin's Minimal Principle), Directed Study, SU2018, SP2019
- t6. Tutorial on Julia, the Computational Programming Language for MEEN grad students, 9/28/2018
- t5. Short Lecture on Bipedal Locomotion with HurToolbox, University of los Andes, Colombia, 5/16/2018
- t4. Short Lecture on Multivariable (Robust and Optimal) Control for MEEN grad students, 5/10/2018
- t3. Crash Course on Dynamics - Newtonian, Lagrangian, Hamiltonian and Kane's Mechanics for MEEN grad students, 1/10/2018
- t2. Crash Course on Robot Opereting System (ROS) for MEEN grad students, 8/19/2016
- t1. Short Lecture and Tutorial on Passive Dynamic Walking for MEEN grad students, 12/12/2014

University of Illinois at Urbana-Champaign *Urbana, IL*

INSTRUCTOR/TEACHING ASSISTANT *Aug 2006 - Dec 2010*

- t4. ME340: Dynamics of Mechanical Systems (Instructor) - SU2009
- t3. ME360: Signal Processing - FA2009
- t2. TAM212: Introduction Dynamics - FA2008, SP2009
- t1. ME460: Industrial Control Systems - FA2006

Advising

Student Advising

POSTDOCTORAL RESEARCHERS

- a3. Kangwoo Lee (Ph.D. from Yonsei University) - Biomechanics of Human Movement, Oct 2023 - Present
- a2. Jinwon Lee (Ph.D. from Ajou University) - Deep learning for bipedal walking, (Currently, Assistant Professor at Gangneung-Wonju National University, Wonju, Korea), Feb 2020 - Jan 2021
- a1. Han U. Yoon (Ph.D. from UIUC) - Haptic customization for training and rehabilitation, (Currently, Assistant Professor at Yonsei University, Wonju, Korea), Sep 2014 - Jun 2017

PH.D. STUDENTS

- a17. Van Thanh Nguyen - Design and control for dynamic 3D bipedal robotic walking Expected to graduate in Feb 2029
- a16. Hyewon Park - Joint impedance estimation Expected to graduate in Feb 2029
- a15. Hyung-Seok Ryu - FEP in robotic control Expected to graduate in Feb 2028
- a14. Sunwoong Moon - Phase variable estimation Expected to graduate in Feb 2028
- a13. Myeongju Cha - Control of transfemoral prosthesis Expected to graduate in Feb 2028
- a12. Kwonseung Cho - Control of lower limb exoskeleton, Expected to graduate in Feb 2027
- a11. Jinsik Ju - Control of robotic manipulators for safety in cyclotron, Expected to graduate in Feb 2027
- a10. Jiyoon Sung - Control of bipedal robots, Expected to graduate in Feb 2028
- a9. Woolim Hong - User- and speed-adaptability enhancement of user gait phase estimation for robotic transfemoral prosthesis control, (Currently, Post doc, North Carolina State University, Raleigh, NC), **Graduated in May, 2022**
- a8. Shawanee Patrick - Improving the design metrics of walking assistive devices, (Currently, Post doc, The Ohio State University, Columbus, OH), **Graduated in Dec 2021**
- a7. Namita Anil Kumar - Towards better user customization of lower-limb assistive devices: data driven control strategies and a self-alignming knee mechanism, (Currently, Senior Robotics and Controls Engineer, Johnson and Johnson, Redwood City, CA), **Graduated in Dec 2021**
- a6. Moein Nazifi - Muscle synergy based rehabilitation for slipping perturbation, (Currently, Postdoctoral Research Scientist at Harvard Medical School), **Graduated in Dec 2020**
- a5. Kenneth Chao - Dynamic bipedal robotic walking with human-like behavior, (Currently, Control Engineer at Flexiv Robotics), **Graduated in May 2019**
- a4. Yitsen Pan - Balance and gait rehabilitation using portable skin stretch feedback, (Currently, Postdoctoral Research Scientist at Georgia Tech), **Graduated in May 2019**
- a3. Christian DeBuys - Control of lower limb exoskeleton for paraplegic patients, Expected to graduate in May 2022
- a2. Dongil Shin - Changed Research Topic with Dr. Palazzolo on Dec 2015
- a1. Yooseok Kim - Stopped Studying due to Personal Reasons on April 2015

M.S. STUDENTS

- a24. Laeheon Kim, GIST, Expected to graduate in Aug 2027
- a23. Suyoung Park, GIST, Expected to graduate in Aug 2027
- a22. Jehyun Park, GIST, Expected to graduate in Feb 2027
- a21. Sunghwan Bae, GIST, Expected to graduate in Aug 2026
- a20. Minseok Kim, GIST, Expected to graduate in Aug 2026
- a19. Moonyoung Yim, Stopped Studying due to Personal Reasons on Jan 2025
- a18. Sunghyun Kim, GIST, Expected to graduate in Aug 2025

a17. Hyewon Park, GIST, changed to direct PhD program in Aug 2024

a16. Hyeon Cho, GIST, **Graduated in Feb 2025**

a15. Jiyoung Jeong, GIST, **Graduated in Aug 2024**

a14. Hyung-Seok Ryu, GIST, **Graduated in Feb 2024**

a13. Myeongju Cha, GIST, **Graduated in Feb 2024**

a12. Sunwoong Moon, GIST, **Graduated in Feb 2024**

a11. Yejin Kang, GIST, Stopped studying due to personal reasons on Oct 2021

a10. Kwonseung Cho, GIST, **Graduated in Aug 2023**

a9. Felipe Constantin Reyes Miftajov, **Graduated in Dec 2020**

a8. Veronica Knisley - Numerical methods for trajectory optimization and control of bipedal walking (Currently at Southwest Research Institute, San Antonio, TX), **Graduated in May 2020**

a7. Yonghee Lee - Human-animal interaction during Hippotherapy, (Currently at LG Innotek, Korea) **Graduated in Dec 2019**

a6. Kenny Chour - Electrotactile display and rehabilitation (Currently, Senior Software Engineer, NASA Ames Research Center, USA), **Graduated in May 2018**

a5. Woolim Hong - Unifying walking control of transfemoral prosthesis for various slopes (Currently, continuing PhD at A&M), **Graduated in Dec 2017**

a4. Namita Anil Kumar - Design of upper limb rehabilitation device for stroke patients using gyroscopic effect (Currently, continuing PhD at A&M), **Graduated in Aug 2017**

a3. Daniel McGowan - Design of hybrid transfemoral prosthesis (Currently, at HX5 at Houston), **Graduated in May 2017**

a2. Shawanee Patrick - Biomechanical analysis of lower limb powered prosthesis (Currently, continuing PhD at A&M), **Graduated in Aug 2016**

a1. Victor Christian Paredes Cauna - Upslope walking with transfemoral prosthesis using optimization based spline generation (Currently, running a startup company, Lima Bionics, in Peru), **Graduated in May 2016**

Undergraduate Students

a35. Choongmin Lee - GIST, Mar 2025 - Current

a34. Sunghwan Kim - GIST, Jan 2024 - Current

a33. Minseok Kim - GIST, Mar 2024 - Current

a32. Hyeonwoo Lee - GIST, Jan 2022 - Current

a31. Leaheon Kim - GIST, Jan 2023 - Aug 2025

a30. Wooyoung Park - GIST, Sep 2022 - Aug 2024

a29. Sunghwan Bae - GIST, Sep 2022 - Aug 2024

a28. Seungyun Lee - GIST, Mar 2022 - Feb 2023

a27. Haewon An - GIST, Jan 2022 - Aug 2024

a26. Juyeong Hong - GIST, Sep 2021 - Aug 2022

a25. Junhyeok Im - GIST, Sep 2021 - Feb 2025

a24. Myoungju Cha - GIST, Jan 2021 - Feb 2022

a23. Sunwoong Moon - GIST, Jan 2021 - Feb 2022

a22. Jordan Jimenez - Aggie Challenge, Sep 2018 - May 2019

a21. Seung Jin Kim - Aggie Challenge, Sep 2018 - May 2019

a20. Chiseung Lee - Aggie Challenge, Sep 2018 - May 2019

a19. Kayoung Lee - Aggie Challenge, Sep 2018 - May 2019

a18. Wookhyun Park - Aggie Challenge, Sep 2018 - May 2019

a17. Katya Anastasia Cope - Aggie Challenge, Jan 2019 - May 2019

a16. Hayoung Jeon - Aggie Challenge, Jan 2019 - May 2019

a15. Patrick Currin - Aggie Challenge, Jan 2019 - May 2019

a14. Ruben Monterrosa - Aggie Challenge, Jan 2019 - May 2019

a13. Reuben Ninan - Aggie Challenge, Jan 2019 - May 2019

a12. Jack Hays - Aggie Challenge, Sep 2018 - Dec 2018

a11. Jae Young Jun - Aggie Challenge, Sep 2018 - Dec 2018
a10. Michael McClure, Sep 2017 - May 2018
a9. Klayton Whittler, Sep 2017 - May 2018
a8. Lanna Lytle, Jan 2015 - May 2018
a7. Tyler Marr - Development of an IMU-based motion capture system, control of a robotic manipulator for the upper limb deficient patients, May 2016 - Aug 2017
a6. Wyatt Hahn - Development of an IMU-based motion capture system, Sep 2016 - May 2017
a5. Seungjun Lee - Validation of the optimal design of an assistive glove for stroke patients, Jan 2017 - May 2017
a4. Ian De Vlaming, Sep 2015 - May 2016
a3. Shyla Escobedo, Jan 2015 - May 2016
a2. Michelle Petersen, Jan 2015 - Dec 2015
a1. Tyler Martin, Jan 2016 - May 2016

Visiting Scholars and Interns

Visiting Scholars

a1. Hak Sung Kim - Associate Professor, Mechanical Engineering, Hanyang University, Seoul, Korea, Apr 2017 - Feb 2018

Interns

a18. Hyunseo Son, Biomedical Engineering, Jeonbuk University, Jan 2025-Feb 2025
a17. Su Yeong Park, Mechanical Engineering, Kookmin University, Jan 2025-Feb 2025
a16. Jehyun Park, Mechanical Engineering, Kookmin University, Dec 2024-Feb 2025
a15. Yongseung Yang, Mechanical Engineering, Kyunghee University, June 2024-Feb 2026
a14. Young Sang Hwang, Smart ICT Convergence, Konkuk University, Jul 2024-Aug 2024
a13. Seung Geon Yoo, Mechanical Engineering, Korea Aerospace University, Jan 2024-Feb 2024
a12. Hyeon Cho, Mechanical Engineering, Kumoh National Institute of Technology, June 2021-Aug 2022
a11. Heonsu Kim, 3D Printable Prosthetic Foot, Hanyang University, Seoul, Korea via global human resource development program for innovative design in robot and engineering, The ministry of trade industry and energy, Korea, Feb 2020-Aug 2020
a10. Huijin Um, Toe Joint Consideration for the Prosthetic Foot, Hanyang University, Korea via global human resource development program for innovative design in robot and engineering, The ministry of trade industry and energy, Korea, Sep 2019-Feb 2020
a9. Seref Yagli, Rehabilitation Robotics, Harmony Science Academy High School, Houston, TX, Research Experiences for Teachers **(RET)** in Mechatronics, Robotics, and Automated System Design, NSF, SU2017
a8. Iván Nicolás Gutiérrez Arias, Biomechanical evaluation of assistive glove design for stroke rehabilitation, Pontificia Universidad Católica de Chile (PUC), SP2017
a7. Stephanie O'Donoghue, Control of series elastic actuators for robot compliance, Chattahoochee Technical College, Acworth, GA, Research Experiences for Teachers **(RET)** in Mechatronics, Robotics, and Automated System Design, NSF, SU2016
a6. Sam Nadell, Development of series elastic actuators for robot compliance, Washington University in St. Louis, St. Louis, MO, Research Experiences for Undergraduates **(REU)** in Mechatronics, Robotics, and Automated System Design, NSF, SU2016
a5. Jose Guillermo Colli Alfaro, Development of virtual rehabilitation environment for stroke patients, Model University, Mexico, **CANIETI** Program (A program for engineering and intensive English Internship to increase the number of Mexican students in graduate programs in the US), SU2016
a4. Cosme Jose Basto Adrian, Development of virtual rehabilitation environment for stroke patients, Superior Tech Institute of Motul, Mexico, **CANIETI** Program, SU2015
a3. Abril Elvira Medina Moreno, Development of virtual rehabilitation environment for stroke patients, Autonomous University of Yucatan, Mexico, **CANIETI** Program, SU2015
a2. Junior Jose Garcia Sosa, Development of virtual rehabilitation environment for stroke patients, Model University, Mexico, **CANIETI** Program, SU2015
a1. Manuel Jurado Ledon, Development of virtual rehabilitation environment for stroke patients, Model University, Mexico, **CANIETI** Program, SU2015

Thesis Committee

Ph.D. Degree

a29. Taehoon Jung (ME, GIST, Jaewook Lee) - Design of Halbach Array In-wheel Motor using Multi-Material Topology Optimization, Defended on Nov 2024

a28. Nam-Jin Park (ME, GIST, Hyosung Ahn) - Controllability and Controllable Subspaces of Structured Networks: A Graph-Theoretical Approach, Defended on May 2024

a27. Donghyun Song (BioE, UNIST, Gwanseob Shin) - Continuous Knee Joint Torque/Angle Prediction from Surface Electromyography during Stair Ambulation:Validation and Application for Assistive Devices, Defended on Dec 2023

a26. Younghun John (ME, GIST, Hyosung Ahn) - Consensus of Multi-agent Systems: Fixed-time Convergence and H∞ Performance, Defended on Nov 2021

a25. Jeonggeun Lim (ME, GIST, Jongho Lee) - Obstacle Magnification for Obstacle Avoidance and Safe Landing Site Detection using Slope Estimation and Semantic Segmentation for a multi-rotor , Defended on Nov 2021

a24. Woolim Hong (MEEN, Pilwon Hur) - User- and speed-adaptability enhancement of user gait phase estimation for robotic transfemoral prosthesis control, Defended on Nov 2021

a23. Shawanee Patrick (MEEN, Pilwon Hur) - Improving the design metrics of walking assistive devices, Defended on Oct 2021

a22. Namita Anil Kumar (MEEN, Pilwon Hur) - Towards better user customization of lower-limb assistive devices: data driven control strategies and a self-alignming knee mechanism, Defended on Oct 2021

a21. Kwon, Seong-Ho (ME, GIST, Hyo-Sung Ahn) - Multi-agent systems: Rigidity based formation control and group consensus, Prelim on May 2021

a20. Quoc Van Tran (ME, GIST, Hyo-Sung Ahn) - Direction-only Distributed Formation Control and Pose Localization in Multi-Agent Systems, Defended on April 2021

a19. Christian DeBuys (MEEN, Pilwon Hur) - Control of lower limb exoskeleton for paraplegic patients, Prelim on Aug 2021

a18. Jon Paul Elizondo (MEEN, Harry Hogan) - Jumping Exercise and Sclerostin Antibody as Countermeasures for Simulated Microgravity in the Adult Rat Skeleton, Defended on March 2021

a17. Muaho Chen (AERO, Robert E. Skelton) - Soft Robotics by Integrating Structure, Materials, Fluids, Control Design, and Signal Processing using the Tensegrity Paradigm, Defended on March 2021

a16. Moein Nazifi (MEEN, Pilwon Hur) - Muscle Synergy for Slip Rehabilitation, Defended on Jun 2020

a15. Chong Ke (ETID, Xingyong Song) - Dynamic modeling and optimization of downhole drilling, Defended on May 2020

a14. Humberto Ramos (AERO, John Hurtado) - Control of flight dynamics, Defended on May 2020

a13. Chi-Wei Kuo (MEEN, Chii-Der Suh) - Controlling bifurcation and dynamic behavior in vibro-impact systems, Defended on Jun 2019

a12. Shao-Chen Hsu (AERO, Raktim Bhattacharya) - Modeling and Control of Mobile Robots Based on Tensegrity Structures, Defended on Jun 2019

a11. Kaimi Gao (MEEN, Bryan Rasmussen) - Dynamic modeling and control architecture design for multi-evaporator pumped two-phase system, Defended on May 2019

a10. Rana Soltani (MEEN, Reza Langari) - Development and control of an active upper limb exoskeleton for rehabilitation, Defended on Jan 2019

a9. Amin Zeiaee (MEEN, Reza Langari) - Development of an upper limb rehabilitation exoskeleton for time-independent coordination training, Defended on Jan 2019

a8. Yitsen Pan (MEEN, Pilwon Hur) - Gait and Balance Rehabilitation with Skin Stretch Feedback, Defended on Dec 2018

a7. Kenneth Chao (MEEN, Pilwon Hur) - Bipedal walking analysis, control and applications towards human-like behavior, Defended on Dec 2018

a6. Serdar Coskun (MEEN, Reza Langari) - Stochastic Decision Making and Advanced-Control Design for an Emergency Lane Change Assistance System in Highway Driving, Defended on Jul 2018

a5. Jaewook Yoo (CSEN, Yoonseok Choe) - Sensorimotor Aspects of Brain Function: Development, Internal Dynamics, and Tool Use, Defended on Feb 2018

a4. Bo Peng (MEEN, Sheng-Jen Hsieh, Pilwon Hur (Co-Chair)) - Modeling and prediction of personalized thermal comfort and control of HVAC system for indoor climate, Prelim on Oct 2017

a3. Ivan De Jesus Diaz Rodriguesz (ECEN, Shankar Bhattacharyya) - Modern design of classical controller, Defended on Feb 2017

a2. Sitae Kim (MEEN, Alan Palazzolo) - Multiple steady state responses prediction for nonlinear rotordynamic systems, Defended on Oct 2016 a1. Chun Lin Yang (MEEN, Chii-Der Suh) - Characterization and Control of Real-World Complex Networks, Planned

M.S. Degree

a34. Hyun Cho (ME, GIST, Pilwon Hur) - Effects of Walking Speed and Visual Stimulus on the Axis Orientation of the Metatarsophalangeal Joint and Foot Progression Angle During Gait, Defended on Dec 2024

a33. Minji Kim (ME, GIST, Kyunghwan Choi) - Impedance Control of Robot Manipulators Without Relying on External Force Information, Defended on Dec 2024

a32. Myeongseok Ryu (ME, GIST, Kyunghwan Choi) - Constrained Optimization-Based Neuro-Adaptive Control (CoNAC) for Euler-Lagrange Systems, Defended on Dec 2024

a31. Jongwon Lee (ME, GIST, Hyosung Ahn) - Control of Airship using Nonlinear MPC and Nonlinear Disturbance Observer, Defended on Dec 2024

a30. Muhammad Umair (ME, GIST, Kyihwan Park) - Hybrid MPC-PI Control for Enhancing Imaging Speed of Non-contact Atomic Force Microscopy, Defended on June 2024

a29. Seunghun Jang (ME, GIST, Kyunghwan Choi) - Online Stator Flux Linkage Estimation of Synchronous Machines, Defended on June 2024

a28. Jiyoung Jeong (ME, GIST, Pilwon Hur) - Environmental perception and joint trajectory generation for transfemoral prosthesis, Defended on June 2024

a27. Gwonyeol Lee (ME, GIST, Seongim Choi) - Adaptive Sampling and Machine learning methods for Efficient Meta-model and application to Real-Time Target Tracking, Defended on Dec 2023

a26. Hyungseok Ryu (ME, GIST, Pilwon Hur) - Estimation of Gait Asymmetry and Gait Speed: Focusing on the Affected Side, Defended on Dec 2023

a25. MyeongJu Cha (ME, GIST, Pilwon Hur) - A Phase Variable Based Phase Estimation for Stair Descent, Defended on Dec 2023

a24. Sunwoong Moon (ME, GIST, Pilwon Hur) - aDMP Based Speed Dependent Gait Pattern Adaptive Phase Variable Estimation, Defended on Dec 2023

a23. Kwonseung Cho (ME, GIST, Pilwon Hur) - Effect of The Toe Joint on Bipedal Walking, Defended on June 2023

a22. Myeonggyun Kim (ME, GIST, Jongho Lee) - Autonomous pursuit unmanned flight system based on location information of a launch-attached location tracker and stereo vision, Defended on June 2023

a21. Joon Oh Kim (ME, GIST, Yong-Gu Lee) - Development and Implementation of Visual Odometry for Autonomous Vehicles, Defended on May 2023

a20. Jaejun Jang (ME, GIST, Seongim Choi) - Data-Driven Real-Time Urban Air Mobility(UAM) Operation Research, Defended on Dec 2022

a19. Kukhee Lee (ME, GIST, Hyunseok Oh) - A Study on Anomaly Detection Using Normalized Autoencoder for the Rotating Machines, Defended on Dec 2021

a18. Yongseok Seo (ME, GIST, Hyunseok Oh) - A Study on Anomaly Detection Using Normalized Autoencoder for the Rotating Machines, Defended on Dec 2021

a17. Veronica Knisley (MEEN, Pilwon Hur) - Numerical methods for trajectory optimization and control of bipedal walking, Defended on March 2020

a16. Yonghee Lee (MEEN, Pilwon Hur) - Exploring how functional improvement is related to interaction between children with cerebral palsy and horses during hippotherapy: a pilot study, Defended on Oct 2019

a15. Kenny Chour (MEEN, Pilwon Hur) - Development of an electrotactile haptic device with application to balance rehabilitation, Defended on Mar 2018

a14. Yalun Wen (MEEN, Prabhakar Pagilla) - Robotic surface finishing of curved surfaces: real-time identification of surface profile and control, Defended on Mar 2018

a13. Aritra Biswas (ASEN, Raktim Bhattacharya) - Dynamics and control of a planar tensegrity robot arm, Oct 2017

a12. Woolim Hong (ECEN, Pilwon Hur, Shankar Bhattacharyya) - Control of powered transfemoral prosthesis, Defended on Oct 2017

a11. Namita Anil Kumar (MEEN, Pilwon Hur) - Hand Rehabilitation Device using Gyroscopic Effects, Defended on Jun 2017

a10. Aditya Vighnesh (ECEN, Aydin Karsilayan) - Analog to Digital converters, Defended on Apr 2017

a9. Daniel McGowan (MEEN, Pilwon Hur) - Design of Hybrid Powered Transfemoral Prosthesis, Defended on Mar 2017

a8. Jacob D. Southern (MEEN, Chii-Der Suh) - High speed spindle design using radial and axial active magnetic bearings, Defended on Mar 2017

a7. Achu G Byju (BMEN, Michael Madigan) - Development of balance recovery training device and biomechanical measures for tripping risk, Defended on Opt 2016 a6. Shawanee Patrick (MEEN, Pilwon Hur) - Biomechanical analysis of lower limb powered prosthesis, Defended on Jun 2016

a5. Victor Paredes (MEEN, Pilwon Hur) - Upslope walking with transfemoral prosthesis using optimization based spline generation, Defended on Mar 2016

a4. Aakar Mehra (MEEN, Aaron Ames) - Analysis of various adaptive cruise controller via experimental implementation, Defended on Jun 2015

a3. Felipe Constantin Reyes Miftajov (MEEN, Pilwon Hur) - Robotic Grinding, Planned

a2. Veronica Knisley (MEEN, Pilwon Hur) - Control of bipedal walking under slipping condition, Planned

a1. Yuan Wei (MEEN, Won-jong Kim) - Control of a motor based on Halbach array, Planned

Service Activities

Gwangju Institute of Science and Technology

DEPARTMENTAL

- Search Committee: SP2021, FA2021, SP2022, FA2022, SP2023, SP2024, FA2024
- Robotics Club Activities Revitalization Task Force: SP2022, FA2022, SP2023
- Curriculum Committee: SP2024, FA2024
- Strategic Development Committee: SP2024, FA2024
- Liaison for the Sangmu Army Infantry School, 2024 - Current
- MRE Day Prep Committee (invited Youtuber “Yak”), 2024
- Academia-Industry Committee Between GIST and LG, 2022 - 2024
- Steering Committee for LG Contract-Based Department, 2024

COLLEGE & UNIVERSITY

- Appointment as Special Ambassador for Science and Technology (Jangseong County), 2024 - Current
- Director of Intelligent Motor Talent Training Center, 2023 - Current
- International Undergraduate Student Admission Committee, 2021
- Domestic Undergraduate Student Admission Committee, 2021 - Current
- Faculty Advisor for University House, 2022 - Current
- University Residential Operations Committee, 2022 - Current
- Faculty Senate, 2022 - 2023
- Operating Committee for the Food Tech Robotics in Jangseong County, 2024
- Motor Boot Camp, SU2004

STUDENT ORGANIZATION

- Advisor for ISAAC (Robotics Club), 2022 - Current

Texas A&M University

DEPARTMENTAL

- Seminar Committee for MEEN 681: FA2014, SP2015, FA2015, SP2016, FA2016, SP2017, FA2017, SP2018
- Qualifier Committee for Control: SP2016 (Alternate member for control area), FA2016 (Member for control area), SP2017 (Chair for control area)
- Distance Learning Committee: FA2016, SP2017, FA2017, SP2018, FA2018, SP2019

COLLEGE & UNIVERSITY

- Committee, Curriculum Development for Equine-Assisted Activities and Therapies, Texas A&M Equine Initiative, April 2019 - Present
- First Generation Engineering (FGEn) Student Mentor, July 2017 - Present
- Engineering & Medicine (EnMed) Thematic Research Subcommittee and Instructor, Oct 2016 - Present
- Explorations Article Reviewer for 2016, 2018

- PESCA Reviewer for 2017, 2018

STUDENT ORGANIZATION
- Advisor for Texas A&M University Robotics Team and Leadership Experience (TURTLE), Jan 2015 - Present
- Advisor for Korean-American Scientists and Engineers Association (KSEA), Aug 2017 - Present

Conferences

ASSOCIATE EDITOR
- IEEE International Workshop on Advanced Robotics (ARSO) and its Social Impacts, Since 2017
- IEEE International Conference on Ubiquitous Robotics (UR), Since 2018
- RAS/EMBS IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), Since 2020
- IEEE/RSJ International Conference on Robotics and Automation (ICRA), Since 2021

REVIEWERS
- World Congress on Biomechanics (WCB)
- American Society of Biomechanics (ASB)
- International Conference on Biomedical Engineering and Biotechnology
- American Society of Mechanical Engineers (ASME) International Mechanical Engineering Congress & Exposition (IMECE)
- ASME Summer Biomechanics, Bioengineering and Biotransport Conference
- ASME International Design Engineering Technical Conferences (IDETC)
- IEEE Haptics Conference
- IEEE Conference on Control Technology and Applications (CCTA)
- IEEE Biomedical Robotics and Biomechatronics
- IEEE Control Systems Society Conference
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Robotics and Automation (ICRA)
- Dynamic Systems and Control Conference (DSCC)
- American Control Conference (ACC)
- Eurohaptics

COMMITTEE
- IFAC TC Member, TC2.4 (Optimal Control), 2023-2026
- IFAC TC Member, TC4.1 (Human Machine Systems), 2023 - 2026
- IFAC TC Member, TC4.3 (Robotics), 2023 - 2026
- ICROS Board Member, 2024 - 2025
- KSME Bioengineering Division Board Member, 2024 - Current
- KSME IT/Convergence Division Board Member, 2022 - Current
- ICROS Elected Council Member, 2023 - 2024
- Vice Chair of International Program Committee for IFAC World Congress 2026
- Program Chair for Institute of Control Robotics and Systems (ICROS) 2026
- Program Committee for ICROS 2025
- Program Committee for ICROS 2024
- Program Committee for ICROS 2023
- Program Committee for ICROS 2021
- ICRA, Human-Robot Interaction in Exoskeletons Session Chair, 2021
- ICRA, Human-Robot Interaction: Medical Robots and Systems I Session Chair, 2021
- IROS, Humanoid and Bipedal Locomotion II Session Chair, 2019
- US-KOREA Conference on Science, Technology and Entrepreneurship (UKC), Robotics and Controls Session Chair, 2018

- IROS, Optimization Session Chair, 2017
- American Society of Biomechanics (ASB), Balance and Sensory Augmentation Session Chair, 2015
- ASB, Falls Session Chair, 2012

Journal

Associate Editor

- International Journal of Control, Automation, and Systems (IJCAS), Since 2018
- Frontiers in Human Neuroscience, Since 2021

Reviewers

- IEEE Access
- IEEE Control System Magazine
- IEEE Transactions on Human-Machine Systems
- IEEE Transactions on Haptics
- IEEE Transaction on Industrial Electronics
- IEEE Transaction on Neural Systems and Rehabilitation Engineering
- IEEE Robotics and Automation Letters
- IEEE/ASME Transactions on Mechatronics
- ASME Journal of Biomechanical Engineering
- Scientific Reports
- International Journal of Intelligent Robotics and Applications (JIRA)
- Journal of Translational Engineering in Health and Medicine
- Advances in Mechanical Engineering
- Journal of Biomechanics
- Journal of Applied Biomechanics
- Journal of Neurophysiology
- Neuropsychologia
- PLOS ONE
- Frontiers in Human Neuroscience
- CRC Press
- Clinical Biomechanics
- Gait and Posture
- Applied Ergonomics
- Annals of Biomedical Engineering
- Medical & Biological Engineering & Computing
- Journal of Engineering in Medicine
- Quality and Reliability Engineering International

Other Services

External Grant Reviewer

- NRF Doctoral Student Research Incentive Grant, Review Panelist, April 2024
- NRF ICT, Sejong Fellowship, Review Panelist, April 2024
- NRF Engineering, Sejong Fellowship, Review Panelist, April 2024
- External Expert Evaluator for the Korea Institute of Machinery and Materials (KIMM), Jan 12, 2024
- NRF Doctoral Student Research Incentive Grant, Review Panelist, July 2023
- NRF Mid-Career Award (Type 1-1), Review Panelist, July 2023
- NRF Basic Research Lab (Multidisciplinary), Review Panelist, April 2023

- NRF ICT, Basic Research, Review Panelist, April 2023
- NRF Regional University, Review Panelist, April 2023
- NRF Engineering, Basic Research, Review Panelist, April 2023
- Evaluator for Jeollanam-do Overseas Student Support Program, April 2023
- Evaluator for Jeollanam-do Overseas Student Support Program, April 2022
- NSF CPS, Review Panelist, June 2019
- NSF CPS, Review Panelist, May 2019
- National Institute of Standards and Technology (NIST) for the US-Israel Binational Industrial Research and Development (BIRD) Foundation, May 2019

External Committee

- The Warrior Platform at the Army Infantry School, Sangmu Military Academy, June 2024 - Current
- Future Business Planning Team, Gwangju Technopark, 2024
- American Society for Testing and Materials (ASTM) F48.02 Committee for standards on Exoskeletons and Exosuits for Industry, Medical and Military Services, Mar 2019 - Current

Professional Membership

- Korea Robotics Society (KROS)
- Institute of Control Robotics and Systems (ICROS)
- Institute of Electrical and Electronics Engineers (IEEE)
- American Control Conference (ACC)
- Wearable Robotics Association
- International Society for Virtual Rehabilitation
- American Heart Association (AHA)
- Society for Neuroscience (SfN)
- World Federation for NeuroRehabilitation
- World Congress on Biomechanics (WCB)
- Korean-American Scientist and Engineers Association (KSEA)
- American Society of Biomechanics (ASB)
- Korean Society of Simulation
- Korean Society of CAD/CAM

Media Coverage

- "The Gwangju Institute of Science and Technology Scientists Confirm Horse Riding as a Viable Mobility Treatment for Cerebral Palsy" *Yahoo Finance*, Oct 29th, 2021
  https://finance.yahoo.com/news/gwangju-institute-science-technology-scientists-123300432.html
- "LIGHTER ROBOTIC TRANSFEMORAL PROSTHESIS DEVELOPED" *The O&P Edge Magazine*, Feb 28th, 2019
  https://opedge.com/Articles/ViewArticle/2019-02-28/lighter-robotic-transfemoral-prosthesis-developed
- "Texas A&M researcher develops lighter robotic prosthesis for amputee patients" *Texas A&M Engineering Experiment Station (TEES)*, Jan 29th, 2019
  https://tees.tamu.edu/news/2019/01/29/texas-am-researcher-develops-lighter-robotic-prosthesis-for-amputee-patients/
- "Invitation for International Seminar of Biomedical Engineering" *Universidad de los Andes*, June 5th, 2018
  https://ingbiomedica.uniandes.edu.co/index.php/es/menu-dpto-news/9-uncategorised/365-cubrimiento-especial-2
- "Researchers developing robotic prosthetics to restore balance" *The O&P Edge Magazine*, May 11th, 2017
  https://opedge.com/Articles/ViewArticle/2017-05-11/researchers-developing-robotic-prostheses-to-restore-balance
- "Researchers developing robotic prosthetics to help restore balance in fall victims" *Texas A&M Engineering*, May 8th, 2017
  https://engineering.tamu.edu/news/2017/05/researchers-developing-robotic-prosthetics-to-help-restore-balance-in-fall-victims.html

- “Engineering partnership touches lives of students, seniors” *University of Wisconsin-Milwaukee*, March 13th, 2014 https://uwm.edu/news/engineering-partnership-touches-lives-of-students-seniors/

Honors & Awards

- IROS Academic Award, ICROS, 2025
- Education Innovation Award, GIST, 2022
- MEEN Research Priming Seed Grant Award, TAMU, 2019
- MEEN Faculty Development Funding, TAMU, 2019
- Horses and Humans Research Foundation Research Award for 2017
- Travel Award, International Conference on Virtual Rehabilitation, 2013
- American Heart Association Postdoc Fellowship, 2012
- Travel Award, World Congress on Biomechanics, 2010
- Paul D. Doolen Scholarship on Aging, Nominated as alternate winner, 2009, 2010
- Graduate Travel Award, UIUC, 2007
- Schaller Travel Award, UIUC, 2007
- National Schoarship, Ministry of Science and Technology, Korea, 2004-2006
- Summa Cum Laude, Mechanical Engineering, Hanyang University, Seoul, Korea, 2004
- Merit-based Scholarship, Hanyang University, Seoul, Korea

Honors & Awards (Students)

- Hyung-Seok Ryu, Best Paper Award, ICROS, 2025
- Sunghwan Bae, Best BS Thesis Award, MRE, GIST, Feb 2025
- Haewon An, Best BS Thesis Award, MRE, GIST, Aug 2024
- Myeongju Cha, Best Paper Award, ICROS, 2024
- Kwonseung Cho, Best Paper Award, KASB, 2023
- Seungyun Lee, Best BS Thesis Award, ME, GIST, Feb 2023
- Sungwoong Moon, Best Paper Award, KASB, 2022
- Juyeong Hong, Best BS Thesis Award, ME, GIST, Aug 2022
- Sungwoong Moon, Best BS Thesis Award, ME, GIST, Feb 2022
- Christian DeBuys, Best Paper Award, SCASB, 2019
- Namita Anil Kumar, Graduate Student Teaching Award, TAMU, 2019
- Moein Nazifi, Teaching Fellow, TAMU, 2019
- Shawanee Patrick, NSF AGEP Texas A&M Research Model Scholarship, 2018
- Moein Nazifi, ASB Travel Grant, 2018
- Yitsen Pan, NATEA-Dallas Scholarship, 2017
- Kenneth Chao, Mechanical Engineering Travel Grant, 2017
- Moein Nazifi, Mechanical Engineering Travel Grant, 2017
- Yitsen Pan, Mechanical Engineering Travel Grant, 2017
- Moein Nazifi, Best Paper Award, SCASB, 2017
- Moein Nazifi, Delsys Travel Grant, 2016
- Han Ul Yoon, TAMU Postdoc Travel Grant, 2016
- Han Ul Yoon, GenDepot Award, 2016
- Woolim Hong, GenDepot Award, 2016
- Moein Nazifi, Mechanical Engineering Travel Grant, 2016

- Shawanee Patrick, ASB Diversity Travel Award, 2016
- Moein Nazifi, Mechanical Engineering Travel Grant, 2015
- Yitsen Pan, Mechanical Engineering Travel Grant, 2015
- Kenneth Chao, Taiwan Government Scholarship for Outstanding Students to Pursue Graduate Study Abroad for 2 years, 2015
- Yitsen Pan, The Association of Chinese American Professionals (ACAP) Student Research Project Contest Finalist, Houston, TX, 2015
- Han Ul Yoon, GenDepot Award, 2014

Outreach & Extra Activities

- Special Lecture for Students from Jangsung High School: Physical AI and Human-Friendly Robots, July 10, 2025
- Special Lecture for Students from Gwangju Science High School: Bipedal Walking Robots: Exoskeletons and Humanoids, July 7-9, 2025
- Special Lecture for Students from Suwan and Jeonnam High Schools: Control of Wearable Robots, Jan 21, 2025
- Lecture on Self-Designed Curriculum at Gwangju Science High School: Applied Mathematics II, Spring 2024
- Judging at the National Gifted Schools Joint R&E Excellence Presentation Conference, Jan 9, 2024
- Judging at the Advanced Independent Research Presentation Contest of Gwangju Science High School, Nov 28, 2023
- Lecture on Self-Designed Curriculum at Gwangju Science High School: Applied Mathematics I, Fall 2023
- Invited Lecture at Gwangju Science Museum’s Science School: Interaction Between Humans and Robots, July 19, 2023
- Special Lecture at Suncheon Jeil High School: Physical Interaction Between Humans and Robots, July 5, 2023
- Lecture on Self-Designed Curriculum at Gwangju Science High School: Applied Mathematics II, Spring 2023
- Training for Mathematics Teacher at Gwangju Science High School: Mathematics Education Using Mathematica, Nov 16, 2022
- Lecture on Self-Designed Curriculum at Gwangju Science High School: Applied Mathematics I, Fall 2022
- Special Lecture at POSCO High School in Incheon: Physical Interaction Between Humans and Robots, May 27, 2022
- Outreach for Society of Women Engineers club, March, 2021, Hereford High School, Parkton, Maryland
- Organizing the Career Development Forum with 150 International Students, and Professionals from Academia and Industry, Sponsored by Korean Government, KSEA, and Houston Consulate, Nov, 3rd, 2018, Texas A&M University
- Organizing 12 TAMU-KSEA Seminars since Fall 2017 on the following Topics: Career Developments for Academia and Industry, Resume Writing, Career Fair Preparation, Successful Life for Undergraduate and Graduate Students, Issues in VISA and Green Card, Field Trip to Companies in Houston
- K-12 Outreach: Discovery Night at Rock Prairie Elementary
- Faculty Mentor for Korean-American Scientist and Engineers Association (KSEA) at TAMU
- Faculty Mentor for First Generation Engineering Students at TAMU (FGEn)
- Helping establish a robotics program in the Department of Electrical and Computer Engineering Technology at Chattahoochee Technical College, Acworth, GA
- Advising a student organization “Texas A&M University Robotics Team and Leadership Experience (TURTLE)”
- Faculty Mentor for Louis Stokes Alliance for Minority Participation (LSAMP) Bridge to Doctorate Program (BTD)
- Robotics club at Deer Creek Intermediate School in Winconsin
- (Ordained) Deacon, Korean Church at Champaign-Urbana
- Web masters for several organizations
- Development of library information system for Korean Language School in Champaign
- Development of Election System for a Korean Church
- Flutist, Piano Accompanist at Korean Churches
- Active runner and cyclist